

Practical1

Question

Develop a C program that demonstrates how Linux executes a command entered by a user:

1. Accept a Linux command as input.
2. Create a child process using `fork()`.
3. Execute the command in the child using `exec()`.
4. Allow the parent to wait using `wait()`.
5. Display the PID of parent and child.

Simple C Program

```
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main() {
    char command[50];

    printf("Enter command: ");
    scanf("%s", command);

    pid_t pid = fork();

    if (pid == 0) {
        // Child process
        printf("Child PID: %d\n", getpid());
        execlp(command, command, NULL);
        printf("Command failed\n");
    }
    else {
        // Parent process
        printf("Parent PID: %d\n", getpid());
        printf("Child PID: %d\n", pid);
        wait(NULL);
        printf("Child finished\n");
    }

    return 0;
}
```

OUTPUT:

```
chinnay_1_vcl@DESKTOP:~$  
Enter command: linux  
Parent PID: 716  
Child PID: 717  
Child PID: 717  
Command failed  
Child finished  
chinnay_1_vcl@DESKTOP:~$
```